

The foreign currency effect in the syndicated loan market of the emerging economies¹

Di Gong

University of International Business and Economics
European Banking Center, Tilburg University

Tao Jiang

University of International Business and Economics

Weixing Wu

University of International Business and Economics

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Abstract: This paper documents a foreign currency effect in the pricing of syndicated loans among emerging economies. That is, syndicated loans denominated in foreign currency are associated with a lower loan spread relative to that denominated in local currency. The results are robust to the inclusion of loan controls, borrower fixed effects or interactions of borrower and year fixed effects. We deal with the endogenous choice of currency by a two-stage procedure and show the results are invariant. In addition, the findings of foreign currency effect are insensitive to adjusting the forward exchange rate. We investigate the causes of the foreign currency effect and show that the syndicate composition, market segmentation and capital account openness affect the results. In the end, we show the findings are pronounced in the financial crisis. We discuss the implications of foreign currency borrowing for real effect and financial stability.

Key words: Foreign currency effect, loan pricing, syndicated loans, emerging economies

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¹ Gong: d.gong@uibe.edu.cn; Tao Jiang: astsachiang@126.com; Weixing Wu: wxwu@uibe.edu.cn. We thank Stephen Sapp (discussant) and Xinyao Huang, and participants at the Symposium of Journal of International Financial Markets, Institutions and Money. This project has been supported by “the Fundamental Research Funds for the Central Universities” in UIBE (15QN03) and “Joint Doctoral Promotion Funds” in UIBE(201501).

1. Introduction

Syndicated loans have been growing fast as a major source of corporate funds in recent years, even for emerging economies. The market volume in emerging economies reached the peak of \$150 billion at the third quarter in 2007 (Chui et al., 2010). Within the emerging market, foreign currency debt is widely prevailing², and a substantially large fraction of syndicated loans are denominated in foreign currency for local borrowers.

This paper examines the pricing of syndicated loans denominated in foreign currency and local currency. In theory, if a firm borrows loans denominated in different currencies, the loan pricing should be identical across various currency denomination, otherwise arbitrage exists. Taylor (1987) offers evidence that covered interest parity (CIP) holds, suggesting that the currency in which a loan is denominated could not be material for loan pricing. In a large sample of syndicated loans from ten emerging economies over the period of 1989-2015, however, we find a foreign currency effect for the borrowers domiciled in emerging economies. In particular, syndicated loans denominated in foreign currency are associated with a lower loan spread relative to that denominated in local currency, after controlling for loan characteristics, year dummies, borrower industry dummies, and borrower country dummies.

This finding is robust to the inclusion of the interaction of borrower country and year fixed effects, borrower fixed effects or the interactions of borrower and year fixed effects. As the currency choice is endogenously determined, we follow Santos and Winton (2008) and adopt a two-stage procedure. In the first stage, we estimate the choice of currency using a Probit model with a vector of macro variables that determine the choice of currency. In the second stage, we use the predicted value of foreign currency (dollar) in the loan spread regressions. Our estimates are largely invariant. In addition, we adjust the loan spreads in various currencies into dollar-equivalent spreads using forward exchange rates and the results hold.

We further analyze the underlying mechanisms of the foreign currency effect. We show that whether the foreign currency is the home currency of the lead banks has explanatory power. Specifically, the interest rates of foreign currency loans are no longer lower if the loan currency is not the home currency of the lead banks. In addition, we find market segmentation is relevant in such a way that foreign currency borrowing at a foreign market reduces the pricing differentials and mitigates the foreign currency effect. Moreover, we find capital account openness also weakens the foreign currency effect. In the end, we show the findings are pronounced in the financial crisis.

² For instance, Mora et al. (2013) and Brown et al. (2011) document the large share of foreign currency borrowing in Lebanon and transition economies, respectively.

Our findings have policy implications. The lower interest rate of foreign currency loans lower the cost of borrowing for local firms, and hence has real effects. Borrower may arbitrage and increase debt in foreign currency. However, the overreliance on foreign currency debt also poses a threat to financial stability of emerging economies. Large scale of foreign currency borrowing is alleged as one key driver for the 1997 Asian financial crisis which started from a currency crisis. When the local currency collapsed, a large amount of firms were unable to repay foreign currency debt and fell into distress.

Our work extends along a number of dimensions of growing literature in global syndicated loan market. Carey and Nini (2007) investigate the pricing discrepancy of syndicated loans in the international markets and provide evidence that, *ceteris paribus*, loans spreads to corporate borrowers are significantly smaller in Europe than in the US. Gaul and Uysal (2013) and Berg et al. (2015) follow this research avenue and offer competing stories to resolve the pricing puzzle. In contrast to these papers focusing on the price differences across syndicated loan markets in advanced economies, we study the price differences across various currencies in emerging economies.

Our study is related to an extant literature on the foreign currency loans. Ivashina et al. (2015) show that Eurozone banks contracted dollar-denominated lending relative to their euro lending in the Eurozone sovereign crisis. Bacchetta and Merrouche (2015) document an increase in foreign currency syndicated loans among leverage Eurozone firms since the financial crisis. These papers, together with Giannetti and Laeven (2012a, 2012b), study the changes in lending volumes of different currencies in US and Europe since the subprime crisis and sovereign debt crisis. By contrast, we take a different look at loan pricing of different currencies in emerging economies.

Our paper also complements a vast of literature on the use of foreign currency loans. Brown et al. (2011) and Mora et al. (2013) study the choice of foreign currency in bank loans in emerging countries, whilst Keloharju and Niskanen (2001), Nandy (2010) and Kedia and Mozumdar (2003) use the sample from advanced economies. Most studies discuss hedging or revenue exposure as reasons to denominate debt in foreign currency. Our paper enriches this literature and shows lower costs of foreign currency borrowing in emerging economies.

Graham and Harvey (2001) show evidence that 44 percent of the firms cite lower borrowing costs as a key reason for issuing foreign currency debt. McBrady and Schill (2007) and McBrady et al. (2010) take the fact that foreign currency debt is cheap as granted and further analyze the firm's choice of currency in debt issuance. Our work complements the literature by examining the reason for lower costs of foreign currency debt. Our paper also enriches the evidence that the CIP failed in emerging syndicated loan markets during the financial crisis period (Du and Schreger, 2015; Du et al., 2016).

The rest of the paper is organized as follows. Section 2 presents the data, empirical strategies and summary statistics. Section 3 shows the baseline regression and a battery of robustness checks. Section 4 discusses plausible channels underlying the foreign currency effect. Section 5 analyzes the foreign currency effect in the financial crisis. Section 6 concludes.

2. Data and empirical strategy

From Thomson Reuter Loan Pricing Corporation's Dealscan, we obtain data on syndicated loans borrowed by nonfinancial corporates in ten emerging economies: Brazil, China, India, Indonesia, Malaysia, Mexico, Philippines, Russia, South Africa and Thailand. Dealscan offers detailed information on loan contracts, such as pricing term, currency domination, syndicated structure and etc. This data set has been widely used in the cross country setting (see, e.g., Giannetti and Laeven, 2012a; Giannetti and Yafeh, 2012). Our sample covers the period from 1989 to 2015. We exclude loans extended to borrowers in financial industries (SIC codes 6000 to 6400, Finance and Insurance). Syndicated loans are usually structured in a number of facilities, also called tranches. We treat multiple facilities in a deal as different loans because spreads, identity of lenders and other contractual features often vary. An individual observation in our analysis is thus a loan facility extended by a syndicate to a borrower.

To identify the foreign currency effect, we employ the following loan pricing model:

$$Loanspread_{i,s,c,t} = \alpha + \beta_i FC_{i,t} + \sum_k \theta_k Loan_{i,t} + \sum_s \xi_s S_s + \sum_c \varphi_c C_c + \sum_t \zeta_t T_t + \varepsilon_{i,s,c,t}$$

where i , s , c and t denote facility, borrower industry, borrower country and year, respectively. The dependent variable, $LoanSpread$, is the all-in-drawn spread over Libor measured in basis points. The all-in-drawn-spread is a measure of the overall costs of the loan, accounting for both one time and recurring fees. FC is a dummy that takes value of one if the denomination of the loan currency is different from the home currency of the borrower, and zero otherwise. Alternatively, we define a dollar dummy in a similar fashion.

In specific, we include a vector of loan controls, $Loan_{i,t}$. First, we include $\text{Log}(\text{FacilitySize})$, the log of the facility amount in million US dollar. Large loans are associated with greater credit risk and lower liquidity in underlying projects, but could also be demanded by larger firms which tend to have lower risks. The relationship between loan size and loan pricing is hence ambiguous. We also include Maturity , the maturity of the facility measured in years. The effect of maturity on loan spreads is also ambiguous, for similar reasons as loan size. Next, to proxy the syndicated structure we use the number of lenders in a facility ($\#Lenders$) and the number of facilities within a deal ($\#Facilities$). To measure the liquidity exposure of each facility, we classify a loan as a line of credit (Revolver) or a term

loan (Termloan). Moreover, we include dummy variables that indicate whether a loan is senior (Senior) in the borrowers' liability structure. Seniority reduces the lenders' exposure in the event of borrower default and therefore lowers lending rates. Despite collateral is crucial for loan pricing, we are unable to restrict to the sample with complete information on collateral because a large fraction of sample lacks such collateral information in Dealscan. We create a dummy (MissingCollateral) for whether the information for collateral is missing, for fear that information may be missing due to any systematic reasons³. Last, we create five dummies to control for the loan purpose: Corporate Purposes (CorpPurpose), Debt Repayment (Repayment), Takeover (Takeover), Working Capital (WorkCapital) and Other (OtherPurposes).

Despite little data on the balance sheet could be matched with the borrowers in emerging markets, we include industry dummies, S_n , that classify borrowers into ten sectors based on one-digit SIC codes, as loss given default (LGD) is strongly correlated with industry characteristics (Hertzel and Officer, 2012; James and Kizilaslan, 2014). In most specifications, we include year dummies, T_t , and borrower country dummies, C_m , to absorb time and country-specific shocks. In some robustness checks, we also include the interaction of borrower country and year fixed effects, borrower fixed effects or the interaction of borrower and year fixed effects. Throughout most specifications we estimate pooled OLS regressions with robust standard errors clustered at the borrower level.

We end up with a sample of 3719 facilities. In our sample, 3181 loans are denominated in foreign currency, and 2997 loans are contracted in US dollar. All-in-drawn spreads are winsorized at the 1 and 99 percent level. Table 1 tabulates the summary statistics.

3. Empirical results

3.1 Baseline regressions

Table 2 reports the output for baseline regressions where all-in-drawn spreads are the dependent variable. We estimate pooled OLS regressions with standard errors clustered at the borrower level. We start by univariate tests in column 1, where we regress loan spreads on FC dummy, controlling for year dummies. The FC dummy obtains a negative coefficient of 223 that is significant at the 1% level. The negative coefficient for the FC dummy variable suggests that the denomination of foreign currency is associated with a lower interest rate. Put differently, loan spreads of loans denominated in foreign currency is lower than that denominated in local currency. Replacing the variable of interests by the dollar dummy in column 2, we find quantitatively and qualitatively similar results. Next, we switch to multivariate analysis with the inclusion of a set of loan controls in columns 3 and 4. Despite a

³ As a large fraction of the sample lacks the information on collateral, unfortunately, data coverage constrains the use of the information on collateral.

smaller size of the estimated coefficients for FC dummy and dollar dummy, they remain statistically significant. We further control for borrowers' industry dummies and country dummies in columns 5 and 6. The estimated coefficients for FC dummy and dollar dummy remain statistically significant with the full set of controls. In addition, a loan denominated in foreign currency would be 166 bps lower than a comparable loan denominated in local currency, which is economically sizable. As regards firm controls, we find that loan spreads are lower if facility size is larger, the facility structure has more lenders, and the loan is a senior debt.

3.2 Robustness checks

In Tables 3 to 5 we report a battery of robustness checks. Throughout all specifications, we include the set of loan controls, but for brevity we no longer report the corresponding estimates.

Despite the inclusion of borrower country fixed effects and year fixed effects, our results might be prone to time-varying macroeconomic shocks, for instance, the change in the interest rate gap between the two countries of local currency and foreign currency. To rule out this possibility, we include a set of borrower country*year fixed effects to take care of these observed and unobserved borrower country-specific time varying shocks. The regressions in the first two columns in Table III further confirm our findings.

In addition, as we have little balance sheet information of the borrowers, the foreign currency effect is likely to be driven by the demand side factors without controlling for borrowers' characteristics. Our results might be problematic if borrowers of different currency denomination differ substantially, particularly in the aspect of creditworthiness. For instance, a borrower of loans denominated in foreign currency might be an exporter and less risky whilst another borrower could be a local manufacturing firm with higher default risk. To address the concern that unobserved borrower heterogeneity drives the pricing pattern of different currency denomination, we exploit the within-borrower variation by including borrower fixed effects. Essentially we are comparing the pricing of loans originated by the same borrower denominated in foreign currency and local currency. Our results in the columns 3 and 4 in Table III are largely the same to that in the baseline regressions. In addition, demand shocks, say, the credit risk of the borrower, may vary over time and hence unobserved time-invariant borrower fixed effects may turn out to be insufficient to control for demand effects. We therefore include borrower*year fixed effects to better take care of any time varying borrower demand shocks. The results in columns 5 and 6 continue to confirm that the lower interest rates associated with foreign currency denomination is not driven by the borrowers' creditworthiness. The deterioration of statistical significance is a natural result of shrinking number of observations within the same borrower in the same year.

Last, our main dependent variable, All-in-Drawn spreads, is a spread over LIBOR measured in basis points. It is noteworthy that the underlying LIBOR varies over time, which cannot be perfectly captured by year dummies. To precisely measure the overall cost for borrowers, we add All-in-Drawn spreads by daily LIBOR and calculate the interest rates of the facility. In the last two columns of Table 3, the results suggest our findings are largely insensitive to the use of interest rates as dependent variable.

As our variables of interest, no matter foreign currency or dollar, are endogenous choice variables determined by the borrower, our results are prone to omitted unobserved factors that drive both loan spreads and choice of currency. Brown et al. (2011) find that the choice of foreign currency is strongly related to the firm's foreign currency revenue. In a hypothetical case, an importing firm may choose foreign currency debt and its business in international trade may affect the risk and hence borrowing costs. To deal with this issue, we follow Santos and Winton (2008) and adopt a two-stage procedure. In the first stage, we estimate the choice of currency using a Probit model. On top of the common set of controls, we further include a vector of macro variables that determine the choice of currency, such as net export to GDP (NetExp), volatility of exchange rates (FXvola), inflation, log GDP and interest rate gap between the borrower country and the US (GAP)⁴. We obtain most country specific variables from World Development Indicators, except volatility of exchange rate from the Wind database. In the second stage, we use the predicted value of foreign currency (dollar) in the loan spread regressions. Columns 1 and 3 in Table IV report the first stage regression, suggesting that volatility of exchange rates and log GDP significantly predict the use of foreign currency and dollar. The estimated currencies are substituted into the second stage regression in columns 2 and 4. The coefficients on the predicted currency choice are negative and statistically significant, continue to supports the findings of the foreign currency effect.

As mentioned in Carey and Nini (2007), the currency in which the loan spread is paid may impact the level of loan spread even if covered interest parity (CIP) holds as the contract spread is an additive constant over Libor. This is particularly true when the local currency in emerging economies have expectations of appreciation or depreciation, especially in the Asian currency crisis in late 1990s. In cases that forward exchange rate implies that the value of local currency will be half relative to the foreign currency in the next year (depreciation of local currency), then we have to convert the local currency spread into foreign currency spread before comparing the two spreads. For instance, a loan of one year maturity paying 100 bps in foreign currency will be equivalent to another loan denominated in local currency paying 200 bps. To address this concern, we convert contract spreads denominated in local currencies as well as foreign currencies into dollar-equivalent spreads, using forward

⁴ Unfortunately we are unable to explicitly control for firm characteristics that affect the choice of currency, which is obviously a drawback.

exchange rates between that currency and US dollar, as of the facility start date. The data on forward exchange rates come from Bloomberg. This in essence guarantees that cash flows in all currencies are converted into a comparable base and hence have the same present values. Despite in certain periods of time the two spreads differ considerably, the dollar-equivalent spread is only 2 bps smaller than the local currency spread on average over the whole sample period. When the dollar-equivalent spread is employed in columns 5 and 6 as the dependent variable in the analogous regressions to the baseline, we continue to find a strong foreign currency effect, suggesting that our results are unlikely driven by the forward exchange rate. The results are similar when adjusting the interest rates as dollar equivalent (see the last two columns).

Lending relationship is likely to reduce loan spreads (Bharath et al., 2011). Our results would be in trouble if relationship loans were dominated in foreign currency and first-time loans were widely denominated in local currency. To address this concern, we split our sample into two sub-groups: relationship lending and first-time lending. Following Bharath et al. (2011), we define relationship lending as loans extended to borrowers that receive a loan from a given lead bank in the five-year look-back period. The summary statistics in Table 1 suggests 68.5% of the sample are relationship lending. The estimates presented in Table 5 show that FC dummy (dollar dummy) is statistically significant in both sub-groups, indicating that the foreign currency effect are independent of lending relationship.

4. What explains the foreign currency effect?

This section considers potential mechanisms for the foreign currency effect in the syndicated loan market of the emerging economies. Our results in Table 6 suggest that home currency of lead banks, location of the market, and capital account openness can explain the foreign currency effect.

4.1 The home currency of lead banks

If the foreign currency is the home currency of the lenders, it is arguably reasonable to observe a lower loan spreads in foreign currency loans. The advantages of lending in home currency for a foreign bank may include the following two points. First, banks usually have plenty and stable funding in home currency, which are presumably cheaper than the funding in emerging economies. The funding advantage of home currency matters for the asset expansion of these foreign banks, especially in financial crisis periods (as shown in Giannetti and Laeven, 2012b; Ivashina et al., 2015; Bacchetta and Merrouche, 2015). Second, by lending in home currency, banks transfer the exchange rate risk to borrowers. The history of currency crisis and turmoil in exchange rates of emerging economies suggest considerable risk embedded in lending local currency. On top of that, the hedge in the exchange rate

forward market might be incomplete for the currencies in emerging economies. Overall, lending in home currency implies plausibly lower funding cost and less exchange rate risk. Hence, if the foreign currency is the home currency of the bank, we expect a lower interest rate.

To test this hypothesis, we define a home currency dummy variable that takes the value of one if the foreign currency is the home currency of all lead banks, and zero otherwise. Following Giannetti and Laeven (2012a), we only consider lead banks as lending banks because other participants in a syndicated loan do not have direct contract with the borrower. It is the lead bank that negotiate the interest rate and non-pricing terms with the borrower on behalf of the syndicate. In addition, we interact the home currency dummy with the foreign currency dummy and present the results in the first two columns of Table 6. The estimated coefficient of FC (dollar) dummy is no longer statistically significant. But the coefficient of the interaction term turns out to be negative and statistically significant, suggesting that foreign currency loans have a lower interest rate only if the lead banks are lending in their home currency. This is in line with our argument of the banks favor loans in their home currency and such loans are associated with lower spreads. In unreported regressions we use the share of lead banks that use their home currency in the syndicate instead of the home currency dummy and find similar results.

4.2 Syndication in foreign markets

Carey and Nini (2007) document the fact that in many cases domiciles of the borrower, investor and transaction of the syndicated deal are quite different and the location market matters for pricing. In our sample, for example, 20 percent of Chinese borrowers sign the syndicate loan contracts in the Hong Kong, and more than 10 percent in the US. Interestingly, 10 percent of syndicated loan borrowed by Malaysian firms are signed in its neighboring country, Thailand. Unlike Carey and Nini (2007) which emphasizes that even for the syndicated loan markets of the US and Europe are not fully integrated, we agree upon that fact the emerging markets are considerably segmented from the advanced markets, both geographically and economically. We, however, believe in our setting that local borrowers are less prone to price discrimination in a foreign market of syndication when borrowing in foreign currency, relative to a disadvantaged position at home. If a local firm borrows foreign currency loans at local market, it is likely to be constrained with a lower bargaining power vis-à-vis foreign banks when the emerging markets are substantially segmented with the international market. Notice that it could be particularly true for foreign currency only.

Therefore we define a foreign market dummy that takes value of one for the loan that is syndicated in the foreign market different from the borrower domicile, and zero otherwise. It is assumed that the market is more segmented if the country of syndication is not the same as

borrower's locality. We interact this dummy with FC (dollar) dummy and expect its coefficient to be positive. The results in columns 3 and 4 present a positive and statistically significant estimated coefficient for the interaction terms, consistent with our hypothesis that borrowing in foreign currency entails a lower interest rate when syndicating in a foreign market relative to that in the local market. In sum, borrowing in a foreign market mitigates the foreign currency effect for the local borrowers in a sizable magnitude.

4.3 Capital account openness

The pricing differential could be a consequence of frictions in capital control in emerging economies. If an emerging economy had a higher degree of capital account openness, the local market would be more integrated with the global market. Hence no-arbitrage condition is more likely to hold, suggesting that the price of various currency denomination should be closer. In sum, we expect that the openness of the capital account may mitigate the foreign currency effect.

In columns 5 and 6, we interact the foreign currency dummy (dollar) with kaopen, an index of degree of capital account openness from Chinn and Ito (2008). Technically, kaopen is the first principal component of the raw variables pertaining to regulatory controls over capital account transactions, the existence of multiple exchange rates, and the requirements of surrendering export proceeds. The estimated coefficient of FC (dollar) dummy remains negative and statistically significant, though much smaller in magnitude. By contrast, the positive and statistically significant coefficient for the interaction term suggests that higher capital account openness raises the interest rates of a FC (dollar) loan relative to a local currency loan, mitigating the foreign currency effect.

5. Financial crisis

The pricing pattern may vary over time, especially over the business cycle. We split the sample into two sub-periods: pre-crisis (1989-2006) and crisis-and-aftermath (2007-2015). The results are reported in Table 7. The estimated coefficient for foreign currency loans is positive prior to the financial crisis, though statistically insignificant, suggesting no evidence of such foreign currency effect in the booming period of syndicated loan markets. The coefficient turns, however, negative and statistically significant in the period of crisis-and-aftermath, in line with our baseline findings. This implies our main findings are basically driven by the crisis period.

6. Conclusions

This paper examines the pricing of syndicated loans denominated in foreign currency and local currency in a large sample of syndicated loans from ten emerging economies over the period of 1989-2015. We find a foreign currency effect that syndicated loans denominated in foreign currency are less costly relative to that denominated in local currency, after controlling for loan characteristics, year dummies, borrower industry dummies, and borrower country dummies. This finding is robust to the inclusion of the interaction of borrower country and year fixed effects, borrower fixed effects, the interactions of borrower and year fixed effects, a two-stage procedure that deals with the endogenous choice of currency, and adjustment of forward exchange rate to the loan spreads.

We further analyze the underlying mechanisms of the foreign currency effect. We show that whether the foreign currency is the home currency of the lead banks has explanatory power. Specifically, the interest rates of foreign currency loan are no longer lower if the loan currency is not the home currency of the lead banks. In addition, we find market segmentation is relevant in such a way that foreign currency borrowing at a foreign market reduces the pricing differentials and mitigates the foreign currency effect. Moreover, we find capital account openness also weakens the foreign currency effect. In the end, we show the findings are pronounced in the financial crisis.

The policy implications are twofold. Many firms in emerging economies build up a large scale of foreign currency debt. On the one hand, cheap foreign currency debt reduces the financial cost for emerging corporate borrowers, benefiting the real economy. On the other hand, the foreign currency debt may increase the financial fragility through the link between exchange rate and corporate balance sheet, posing systemic risk to the emerging economies.

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Table 1 Summary statistics

Variable	Definition	obs	mean	sd	Source
AllinDrawn	All-in-Drawn spreads over LIBOR	3719	263.856	275.883	Dealscan
FC	Dummy variable that equals one if the loan dominated in foreign currency, and zero otherwise	3719	0.855	0.352	Dealscan
dollar	Dummy variable that equals one if the loan dominated in US dollar, and zero otherwise	3535	0.848	0.359	Dealscan
Log(FacilitySize)	Logarithm of the face amount of the loan(in million US dollar)	3719	4.242	2.112	Dealscan
Maturity	Maturity of loan contract in years.	3719	5.089	3.357	Dealscan
#Lenders	Number of the lenders in syndicated loan	3719	8.421	6.996	Dealscan
#Facilities	Number of the facilities in each package	3719	1.982	1.675	Dealscan
Revolver	Dummy variable that equals one if specific tranche type is revolver, and zero otherwise	3719	0.136	0.342	Dealscan
Termloan	Dummy variable that equals one if the syndicated loan is term loan, and zero otherwise	3719	0.783	0.412	Dealscan
Senior	Dummy variable that equals one if the syndicated loan is senior debt, and zero otherwise	3719	0.996	0.061	Dealscan
MissingSecured	Dummy variable that equals one if missing information on the collateral of the loan contract, and zero otherwise	3719	0.656	0.475	Dealscan
CorpPurpose	Loan purpose dummy indicates loans borrowed for corporate purposes.	3719	0.270	0.444	Dealscan
Repayment	Loan purpose dummy indicates loans borrowed for debt repayment.	3719	0.436	0.496	Dealscan
Takeover	Loan purpose dummy indicates loans borrowed for takeover.	3719	0.197	0.398	Dealscan
WorkCapital	Loan purpose dummy indicates loans borrowed for net working capital.	3719	0.031	0.173	Dealscan

OtherPurposes	Loan purpose dummy indicates loans borrowed for purposes other than the previous four.	3719	0.066	0.248	Dealscan
ForeignMarket	Dummy variable that equals one if the country of syndication differs from the borrower domicile, and zero otherwise.	3719	0.114	0.318	Dealscan
HC	Home currency dummy that takes value of one if all lead banks use their home currency, and zero otherwise.	3719	0.195	0.396	Dealscan
kaopen	An index that measures the extent of the openness in capital account transactions.	3431	-0.344	0.942	(Chinn and Ito,2008)
NetExp	Net exports of goods and services (% of GDP).	3688	-20.287	17.681	World Development Indicators
FXvola	Standard deviation of daily exchange rate of a borrower's home country currency, normalized by $\sqrt{252}$.	3614	624.820	2677.798	Wind
Inflation	Measured by the consumer price index.	3688	7.501	8.339	World Development Indicators
Log GDP	Log GDP at market prices (current US\$).	3688	27.691	1.064	World Development Indicators
GAP	A gap of lending interest rates between borrower's home country and the US in percentage.	3663	3.208	13.331	World Development Indicators
REL	Dummy variable that equals one if a borrower receives a loan from a lead bank in the five-year look-back period, and zero otherwise.	3719	0.685	0.465	Dealscan
Interest_rates	Add daily dollar LIBOR to the All-in-Drawn spreads (bp).	3524	548.573	310.581	Dealscan and Wind

Table 2 Baseline results

The dependent variable in all regressions is all-in-drawn spreads. Time fixed effects are included throughout all specifications. Robust standard errors are clustered at the borrower level. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
FC	-223.037*** (23.908)		-155.264*** (28.487)		-165.000*** (27.151)	
dollar		-224.366*** (23.945)		-140.096*** (29.820)		-156.956*** (28.610)
Log(FacilitySize)			-13.498** (5.424)	-17.389*** (6.005)	-9.671* (5.609)	-12.207* (6.292)
Maturity			-0.076 (1.844)	0.299 (1.875)	0.736 (1.798)	0.944 (1.838)
#Lenders			-2.442*** (0.848)	-2.264** (0.907)	-3.041*** (0.781)	-2.964*** (0.843)
#Facilities			-2.645 (4.499)	-7.295** (3.576)	1.981 (3.824)	-2.587 (3.215)
Revolver			-28.297* (17.116)	-22.690 (15.682)	-16.703 (17.297)	-6.803 (15.946)
Termloan			8.979 (15.967)	16.380 (14.473)	18.730 (15.487)	29.041** (14.188)
Senior			-192.539** (82.465)	-201.783** (89.172)	-207.400** (80.964)	-217.601** (87.191)
MissingSecured			-7.980 (9.931)	-10.022 (10.236)	10.850 (10.616)	10.241 (11.045)
_cons	459.711*** (41.291)	460.911*** (41.313)	639.384*** (93.508)	651.371*** (98.630)	443.167*** (93.339)	446.833*** (98.397)
Purpose Dummy	NO	NO	YES	YES	YES	YES
Industry Dummy	NO	NO	NO	NO	YES	YES

Year Dummy	YES	YES	YES	YES	YES	YES
Borrower Country Dummy	NO	NO	NO	NO	YES	YES
Observations	3719	3535	3719	3535	3719	3535
Adj. R-squared	0.205	0.206	0.234	0.240	0.295	0.300

Table 3 Robustness checks

The dependent variable in all regressions is all-in-drawn spreads. Columns 1 and 2 include the interactions between borrower country dummy with year dummy. Columns 3 and 4 run a panel regression with borrower FE. Columns 5 and 6 run a panel regression with borrower and year FE. Robust standard errors are clustered at the borrower level. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Borrower country*Year FE		Borrower FE		Borrower*Year FE		Interest rates	
FC	-150.894*** (27.026)		-133.060** (52.503)		-345.020** (157.077)		-161.666*** (28.545)	
dollar		-148.929*** (28.970)		-125.609** (54.869)		-308.059* (164.264)		-153.375*** (30.209)
_cons	574.214*** (86.116)	569.211*** (90.828)	470.036*** (90.098)	475.080*** (95.869)	661.211*** (149.640)	631.924*** (155.531)	452.824*** (93.101)	457.472*** (98.003)
Loan controls	YES	YES	YES	YES	YES	YES	YES	YES
Purpose Dummy	YES	YES	YES	YES	YES	YES	YES	YES
Industry Dummy	YES	YES	NO	NO	NO	NO	YES	YES
Year Dummy	NO	NO	YES	YES	NO	NO	YES	YES
Country*Year FE	YES	YES	NO	NO	NO	NO	NO	NO
Borrower FE	NO	NO	YES	YES	NO	NO	NO	NO
Borrower*Year FE	NO	NO	NO	NO	YES	YES	NO	NO
Observations	3719	3535	3719	3535	3719	3535	3524	3344
Adjusted R-squared	0.349	0.350	0.154	0.157	0.123	0.107	0.294	0.300

Table 4 Robustness checks continue

Column 1 and 3 estimate the first-stage regression using Probit models, where the dependent variables are a dummy for foreign currency loan and local currency loan, respectively. In the first-stage, are estimated with robust standard errors and clustered by firm. Column 2 and 4 estimate the second-stage regression using OLS where the dependent variable is all in drawn spreads. PFC and Pdollar are predicted FC dummy and predicted dollar dummy, respectively. In Columns 5 and 6, the dependent variables are dollar-equivalent spreads. In Columns 7 and 8, the dependent variables are dollar-equivalent interest rates. Robust standard errors are clustered at the borrower level. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

	(1) FC	(2) AllinDrawn	(3) dollar	(4) AllinDrawn	(5) dollar-equivalent spreads	(6) dollar-equivalent spreads	(7) dollar-equivalent interest rates	(8) dollar-equivalent interest rates
NetExp	0.009 (0.006)		0.015** (0.007)					
FXvola	0.000*** (0.000)		0.000*** (0.000)					
Inflation	-0.022 (0.019)		-0.014 (0.023)					
Log GDP	0.153* (0.088)		0.187* (0.099)					
GAP	-0.002 (0.008)		-0.002 (0.009)					
PFC		-193.169*** (48.594)						
Pdollar				-152.465*** (43.692)				
FC					-149.992*** (28.001)		-149.924*** (29.716)	
dollar						-141.622*** (29.516)		-141.390*** (31.419)
_cons	-4.533 (2.863)	510.208*** (102.693)	-6.304** (3.119)	476.764*** (101.295)	441.685*** (93.674)	447.117*** (98.708)	506.156*** (101.790)	509.373*** (107.425)
Loan controls	YES	YES	YES	YES	YES	YES	YES	YES

Purpose Dummy	YES	YES	YES	YES	YES	YES	YES	YES
Industry Dummy	YES	YES	YES	YES	YES	YES	YES	YES
Year Dummy	YES	YES	YES	YES	YES	YES	YES	YES
Country Dummy	YES	YES	YES	YES	YES	YES	YES	YES
Observations	3460	3460	3279	3460	3662	3478	3503	3323
Adj. R-squared		0.294		0.293	0.288	0.293	0.483	0.488

Table 5 Lending relationship

The dependent variable in all regressions is all-in-drawn spreads. Columns 1 and 2 estimate loans to first-time borrowers only. Columns 3 and 4 estimate loans extended to borrowers that have received loans from the lead banks in the five-year look-back period. Robust standard errors are clustered at the borrower level. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	First-time Lending		Relationship Lending	
FC	-189.700*** (34.922)		-124.227*** (41.108)	
dollar		-183.218*** (36.688)		-119.423*** (42.950)
Constant	386.312*** (146.311)	336.787** (138.297)	497.771*** (114.228)	529.404*** (121.454)
Loan controls	YES	YES	YES	YES
Purpose Dummy	YES	YES	YES	YES
Industry Dummy	YES	YES	YES	YES
Year Dummy	YES	YES	YES	YES
Country Dummy	YES	YES	YES	YES
Observations	1171	1107	2548	2428
Adjusted R-squared	0.294	0.295	0.320	0.324

Table 6 What explains the foreign currency effect?

The dependent variable in all regressions is all-in-drawn spreads. Columns 1 and 2 estimate the effect of home currency. Columns 3 and 4 estimate the effect of location of markets. Columns 5 and 6 estimate the impact of capital account openness on the foreign currency effect. Robust standard errors are clustered at the borrower level. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	Home currency		Location of market		Capital account openness	
FC	-7.404 (28.840)		-171.443*** (27.841)		-69.256*** (26.529)	
FC*HC	-201.415*** (37.792)					
HC	235.067*** (35.898)	231.682*** (36.217)				
dollar		-7.803 (31.039)		-162.586*** (29.403)		-57.129** (27.834)
dollar*HC		-190.186*** (38.489)				
FC*ForeignMarket			159.915** (66.711)			
ForeignMarket			-153.480** (67.048)	-150.467** (67.072)		
dollar*ForeignMarket				152.348** (66.681)		
FC*kaopen					137.448*** (24.282)	
kaopen					-149.487*** (22.887)	-150.513*** (22.959)
dollar*kaopen						140.304*** (24.276)
_cons	408.651*** (86.982)	407.479*** (91.463)	444.838*** (93.471)	448.195*** (98.542)	381.889*** (119.872)	400.739*** (126.487)

Loan controls	YES	YES	YES	YES	YES	YES
Purpose Dummy	YES	YES	YES	YES	YES	YES
Industry Dummy	YES	YES	YES	YES	YES	YES
Year Dummy	YES	YES	YES	YES	YES	YES
Country Dummy	YES	YES	YES	YES	YES	YES
Observations	3719	3535	3719	3535	3431	3266
Adj. R-squared	0.370	0.372	0.296	0.301	0.338	0.343

Table 7 Financial crisis

The dependent variable in all regressions is all-in-drawn spreads. Columns 1 and 2 estimate over the period prior to the financial crisis. Columns 3 and 4 estimate the period of financial crisis and its aftermath. Robust standard errors are clustered at the borrower level. ***, **, * denote significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)
		pre-crisis 1989-2006	crisis-and- 2007-2015	aftermath
FC	12.530 (29.870)		-186.916*** (32.553)	
dollar		17.790 (31.223)		-182.530*** (34.071)
_cons	243.293 (158.713)	252.011 (158.031)	446.824*** (110.398)	446.716*** (119.001)
Loan controls	YES	YES	YES	YES
Purpose Dummy	YES	YES	YES	YES
Industry Dummy	YES	YES	YES	YES
Year Dummy	YES	YES	YES	YES
Country Dummy	YES	YES	YES	YES
Observations	1463	1402	2256	2133
Adj. R-squared	0.326	0.341	0.274	0.277